

Tide retrospective: *cross-susceptibility derivative vanishes for the harmonic Gibbs*

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Abstract

A six-line Tide-loop excursion in the `laplace` seabed, composing two prior tides into a downstream theorem neither delivers alone: $\partial_h \text{Cov}_h(x, x)|_{h=0} = -t \cdot \kappa_3(x, x, x) = 0$ for the harmonic Gibbs $L = (\lambda/2)x^2$ with linear perturbation $A = x$. Lives in `Laplace/OneD/HarmonicCrossSuscDeriv.lean` (87 lines, including header docs); the proof body is six lines on top of `harmonic_id_gibbsRegularity` (Tide 10), `kappa3_harmonic_id_id_id_eq_zero` (Tide 5), and threepoint's generic cross-susceptibility identity (Tide 6). Zero `sorry`, zero `axiom`, zero `native_decide`. The interesting friction in this run was operational, not mathematical: two other Tide agents (separable-potential and bounded-prior-test-function) were operating concurrently in the same checkout, and the `Laplace.lean` import line and the new file itself were churned by branch-switching and staging interleavings, including a commit that landed my file content under another agent's draft commit message. Resolution: amend the bad message; carry the *per-Tide git worktree* recommendation forward (it had already been raised by the 2D pure-quartic retrospective on 2 May).

1 Setting

The threepoint paper develops the susceptibility framework around three load-bearing identities: the FDT ($\partial_h \langle \varphi \rangle_h = -t \text{Cov}(\varphi, A)$), the cross-susceptibility identity ($\partial_h \text{Cov}_h(\varphi, B) = -t \kappa_3(\varphi, A, B)$), and the four-point flow equation. The threepoint repo (Tide 6 + earlier) proves these in the generic-Gibbs setting, parameterised by a `GibbsRegularity` structure and per-observable `GibbsObservable` hypotheses.

For the framework to deliver concrete theorems on a specific potential, a project needs (i) the `GibbsRegularity` instance for that potential, and (ii) `GibbsObservable` instances for the observables of interest. Tide 10 supplied (i) for the harmonic + linear case (`Threepoint.harmonic_id_gibbsRegularity`), and the Tide 10 retrospective named the natural one-tide capstone:

GPT-5.5 Pro: Cross-susceptibility derivative theorems for the harmonic case. With the regularity instance now in hand, the FDT and cross-susceptibility derivatives specialise concretely. The cross-susceptibility derivative is particularly satisfying: by Tide 5, $\kappa_3(\varphi, x, B) = 0$ for the harmonic case, so $\partial_h \text{Cov}_h(\varphi, B)|_{h=0} = -t \cdot \kappa_3(\varphi, x, B) = 0$ for any φ, B for which the relevant `GibbsObservable` hypotheses hold. A natural one-tide step would package this as a stand-alone theorem in `laplace`.

The candidate (G1 in the May 6 survey) is that capstone. The Tide-pick auto-proceed timer fired on G1; this Tide implements it.

The Step 2 deliberation¹ caught two issues with the survey’s wording and an alternative route worth recording.

2 The thing formalised

*For the harmonic Gibbs measure $\mu_{t,h}(x) \propto \exp(-t((\lambda/2)x^2 + hx))$ on $\mathbb{R}$ against Lebesgue, with $\lambda, t > 0$, and given *GibbsObservable* hypotheses for the monomials x , x^2 , and x^3 at the harmonic + linear setup,*

$$\left. \frac{\partial}{\partial h} \text{Cov}_h(x, x) \right|_{h=0} = 0.$$

The Lean signature, in namespace `Laplace.OneD`:

```
theorem cov_h_id_id_deriv_harmonic_eq_zero
  {lam t : ℝ} (hlam : 0 < lam) (ht : 0 < t)
  (hx  : Threepoint.GibbsObservable (volume : Measure ℝ)
    ((λ/2).2) (id) t (id))
  (hx2 : Threepoint.GibbsObservable (volume : Measure ℝ)
    ((λ/2).2) (id) t (id*id))
  (hx3 : Threepoint.GibbsObservable (volume : Measure ℝ)
    ((λ/2).2) (id) t (id*id*id)) :
  HasDerivAt
    (fun h => Threepoint.gibbsCov (volume) ((λ/2).2) (id) t h
      (id) (id))
    0 0
  (using      (λ/2).2      as      shorthand      for      fun x : ℝ => lam / 2 * x
2andidforfun x : ℝ => x; the filesourcespellstheseout).
```

The hypothesis pattern is that the three external *GibbsObservables* encode the differentiation-under-the-integral content for the three monomials. Closing them is candidate G4 of the May 6 survey, deferred here on grounds of scope.

The conclusion is in fact weaker than what holds. By square completion, the perturbed Gibbs measure factors as $\exp(th^2/(2\lambda)) \cdot$ (Gaussian centred at $-h/\lambda$ with variance $1/(\lambda t)$), so $\text{Cov}_h(x, x) = 1/(\lambda t)$ is *globally* constant in h . The κ_3 -route delivers only the basepoint version of this fact; the global-constancy form is recorded as a future tide.

3 Proof strategy

The proof is a structural composition of three prior results, with ring-level arithmetic at the joins. Five lines of Lean tactics:

1. Discharge *GibbsRegularity* via Tide 10’s instance.
2. Apply threepoint’s `gibbsCov.deriv_eq_neg_t_kappa3` with (φ, A, B) =
 $(\text{id}, \text{id}, \text{id})$. The six *GibbsObservable* slots $(\varphi, B, \varphi B, \varphi A, BA, \varphi BA)$

¹See `projects/patterning/tide-log/2026-05-06-tide-cross-susc-deriv-harmonic.md`.

collapse to three distinct hypotheses (hx fills φ and B ; hx2 fills the three pairwise products; hx3 fills the triple). The conclusion is HasDerivAt of the covariance, with target derivative $-t \cdot \kappa_3(\text{id}, \text{id}, \text{id})$.

3. Rewrite the κ_3 term using Tide 5 (Threepoint.kappa3_harmonic_id_id_id_eq_zero) to get $-t \cdot 0$.
4. mul_zero reduces to 0.
5. exact the resulting hypothesis.

The interesting structural feature is what does *not* appear: no calculus done by hand, no integration, no differentiation under the integral, no parity argument carried out in this file. Tide 5 is where the parity argument lives; Tide 10 is where the analytic content of the partition function lives; this tide is just the gate that admits both.

4 Roadblocks and resolutions

The retrospective’s claim about Tide 5 was loose. The Tide 10 retrospective (which this Tide is the capstone of) says ‘‘by Tide 5, $\kappa_3(\varphi, x, B) = 0$ for the harmonic case’’, suggesting Tide 5 covers general (φ, x, B) . It does not. Tide 5 covers only (id,id,id). GPT-5.5 Pro caught this immediately at the deliberation step and supplied a concrete Gaussian counterexample for the general claim:

GPT-5.5 Pro: For a centred Gaussian with variance σ^2 , $\kappa_3(x^2, x, x) = \mathbb{E}[x^4] - \mathbb{E}[x^2]\mathbb{E}[x^2] = 3\sigma^4 - \sigma^4 = 2\sigma^4 \neq 0$. So the honest G1 really is the restricted $(\varphi, B) = (\text{id}, \text{id})$ version.

The deliberation log records this counterexample so the next tide that wants the general (φ, x, B) statement starts from the right premise.

No proof-side roadblocks beyond a whitespace warning. The six-line proof landed first try modulo a single linter warning (an extra space before a colon in a hypothesis declaration). Lean’s elaborator handled the integrand unification at the GibbsObservable slots automatically: the β -equivalence between $(\text{fun } w \mapsto (\text{fun } x \mapsto x)w \cdot (\text{fun } x \mapsto x)w)$ and $(\text{fun } x \mapsto x \cdot x)$ went through without a simp or show; GPT had warned about possible parenthesisation friction, which did not materialise.

Concurrent-Tide-agent collisions. The dominant friction in this run was operational, not mathematical. Two other Tide branches were running in the same checkout during the same window. Concrete symptoms observed:

- *Import-line clobber.* My one-line addition to Laplace.lean (the import for the new file) was overwritten within the staging window: by the time the build started, another agent had replaced my line with their own QuarticBoundedPriorTestFn import. Re-add fixed it.

- *File disappearance on branch-switch.* The new file² was untracked at the moment when the working tree was switched between branches by another agent; after the switch, the file was gone from disk. Switching back to my branch restored it (because the file was actually re-tracked on it under a stray commit).
- *Stray commit with wrong message.* A commit landed on this Tide's branch³ containing exactly my work (the new file plus a one-line Laplace.lean addition, 88 lines total) but with the bounded-prior agent's commit message. Resolution: `git commit --amend` to fix the message; the content was already correct. Reflog has the original.

The 2D pure-quartic retrospective (Tide step 5 in the laplace chain, 2 May) had already raised exactly this category of failure and named the structural fix:

GPT-5.5 Pro: Concurrent Tide agents need filesystem isolation. The Laplace.lean import-list is a contention point because every new top-level file edits it. Two agents in the same checkout will race; it is not a question of careful sequencing. The structural fix is to run each Tide agent in its own git worktree attached to its own branch.

Today's run is one more data point. The fix is overdue.

5 What was Mathlib, what was new

Mathlib provided. Nothing except `mul_zero`. The proof body is purely ring-level / rw-level manipulation of an existing `HasDerivAt` hypothesis.

Threepoint and **laplace** provided. All the infrastructure. The three load-bearing prior tides: `Threepoint.harmonic_id_gibbsRegularity` (Tide 10), `Threepoint.kappa3.harmonic_id_id_id_eq_zero` (Tide 5), `Threepoint.gibbsCov_deriv_eq_neg_t_kappa3` (Tide 6, the generic cross-susceptibility identity).

What was new. The *composition*. A theorem that combines three pre-existing pieces and unconditionally produces a named harmonic-case fact, with the three remaining `GibbsObservable` hypotheses left as parameters. Not novel mathematical content; novel *packaging* that turns a parameterised lemma into a theorem about a specific Gibbs measure. This is exactly the bridge step the Tide loop is good at producing.

6 Lessons

- *Concurrent Tide agents in the same checkout remain a structural failure mode.* The 2 May retrospective named the fix (`git worktree` per Tide branch);

²`HarmonicCrossSuscDeriv.lean`.

³`tide/cross-susc-deriv-harmonic`.

today’s run is an overdue reminder. Worth lifting into the Tide skill as a preflight step that automatically creates a worktree, not just as advice.

- *Retrospectives leak loose claims.* The Tide 10 retrospective overstated Tide 5’s content ($\kappa_3(\varphi, x, B) = 0$ for general (φ, B)). A future tide that read the retrospective and not the source might have wasted time on a false target. Discipline: when a retrospective references a prior tide’s theorem, copy the *Lean signature verbatim*, not a paraphrase. The deliberation step caught this here, but only because GPT supplied a counterexample; without the multi-model loop the loose claim might have propagated.
- *Composition tides are honest tide steps.* A six-line proof is a perfectly valid Tide outcome, provided the deliberation step does the work of identifying the precise composition shape and the right premises. The package-of-three-premises design here makes the theorem’s place in the ecosystem unambiguous: anyone who later discharges the three GibbsObservable hypotheses (candidate G4) gets the unconditional version in one extra line.
- *The numerical sanity check still earned its keep.* For a “structural” identity ($\partial_h \text{Cov} = 0$) it would have been tempting to skip the check. Doing it surfaced the stronger fact that Cov_h is *globally* constant ($1/(\lambda t)$), not just locally constant at the basepoint. That observation became the runner-up alternative GPT named, recorded as a future tide candidate. A skipped check would have left it unnoticed.

7 Follow-ups

- *G4: discharge the three GibbsObservable hypotheses.* For each $k \in \{1, 2, 3\}$, prove $\text{Threepoint.GibbsObservable}(\text{volume})((\lambda/2)^2)(\text{id})t(\text{fun } x \mapsto x^k)$. Closed-form route via square completion + Gaussian moments (per GPT’s recommended derivation). Estimated ~ 300 -- 400 lines for the three cases together; makes today’s theorem unconditional.
- *Global-constancy strengthening (the runner-up).* Prove $\text{Threepoint.gibbsCov}(\text{volume})((\lambda/2)^2)(\text{id})th(\text{id})(\text{id}) = 1/(\lambda t)$ for all h , not just locally constant at $h = 0$. Direct via shifted-Gaussian translation invariance plus the closed-form partition already in Tide 10. Avoids GibbsObservable hypotheses entirely. GPT’s runner-up suggestion at deliberation; not chosen for this Tide because the package shape was different.
- *Structural fix for concurrent-Tide-agents.* A wrapper that runs git worktree add per Tide branch and points the agent at the worktree. Doing this once removes the entire class of collisions documented above. Not a Lean tide; a process tide for the Tide skill itself.
- *Update the May 6 candidates survey.* Mark G1 as Closed. The skill’s Step 4 close-out section requires this.

- `\leanref-tag the threepoint paper`. The harmonic case now has three formalised theorems (`kappa3_harmonic_id_id_id_eq_zero`, the `GibbsRegularity` instance, and today's `cov_h_id_id_deriv_harmonic_eq_zero`), enough for a one-pass tagging consistent with the laplace primer setup. This is candidate H4 in the May 6 survey.

Acknowledgements

GPT-5.5 Pro contributed the deliberation-stage advisory at Step 2, including the explicit Gaussian counterexample to the general $\kappa_3(\varphi, x, B) = 0$ claim and the runner-up shifted-Gaussian alternative. The full consult is preserved at `projects/patterning/tide-log/gpt55_tide_cross_susc_deriv_harmonic_v1.md`.